

ICT80004 Weekly Communication – Week #07

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Industry Supervisor: Dr. Shigang Liu

Date Prepared: 19/09/2025 Internship Week #: 7

Day	Date	Task(s) completed
1	Monday 15 Sep 2025 8 hours	- Finalised the prompt attack test suite grid: Completed direct vs. paraphrased prompt coverage for <i>Violence & Hate, Guns & Illegal Weapons, and Criminal Planning</i> categories. - Mapped each case across 10 social-engineering techniques (Urgency, Persuasion, Incentive, Quid-Pro-Quo, Foot-in-the-Door, Trusted Relationship, Impersonation, Pretexting, Attention Grabbing, Visual/Verbal Deception). - Populated results into the Dataset Grid with predicted outcomes, aligned with empirical behaviour seen in earlier testing. - Consolidated results with screenshots and annotations for systematic evaluation.
2	Wednesday 17 Sep 2025 8 hours	- Reviewed the paper “<i>Trust Me, I Know This Function: Hijacking LLM Static Analysis using Bias</i>”. Key learning: LLMs show abstraction bias — they shortcut reasoning on familiar code patterns, missing subtle bugs. - This leads to Familiar Pattern Attacks (FPAs), where attackers insert small but deterministic perturbations that mislead the LLM while leaving runtime behaviour unchanged. - FPAs are transferable across models and languages, similar to the way indirect prompt injection patterns transfer across LLM families. - This complements my vulnerable code testing (Weeks 3–5), showing that <i>both adversarial prompts and adversarial code patterns</i> can exploit structural biases in LLMs.

Total hours completed for the week: 16

Plans for next week: #08 week (22 - 26 Sept 2025) —————

- Expand beyond current dataset by **probing more vulnerabilities**: Explore multi-turn conversational steering, layered pretexting, and reasoning-model bypasses.
- Investigate connections between **code-based attacks (FPAs)** and **prompt-based injection attacks** for a unified taxonomy.
- Begin drafting **comparative evaluation** of model guardrails (GPT-5, GPT-4, Gemini, Claude, Grok, Llama, DeepSeek).
- Prepare summary slides linking: (a) vulnerable code suite, (b) prompt injection suite, and (c) literature (incl. "Trust Me, I Know This Function" paper).

Screenshot of Timely EMAIL communication update to the Supervisor at the end of week #07 sent on 19 September 2025 Friday at 4:21 PM (16:21).

